
**Team Name : Trendy Brains**

Team Leader : Prateek Apurva

Problem Statement : Redrob — Intelligent Candidate Discovery & Ranking Challenge

# Solution Overview

## ✕ Traditional Approach

**Keyword matching on skills section**

Candidate lists "RAG, LangChain, GPT-4"
→ Keyword filter triggers → shortlisted
→ But they never built anything real

**Why this fails:**
- Skills section is trivially gamed
- Misses people who built real systems but didn't use the "right" keywords
- No evidence validation at all
- Cannot detect honeypot profiles

VS

## ✓ Resume Shotlister

**Reads what candidates actually built**

- Embeds career history (not skills section)
- FAISS semantic search: narrative vs JD
- LLM verifies real work evidence (boolean)
- 5-rule honeypot detector eliminates fakes

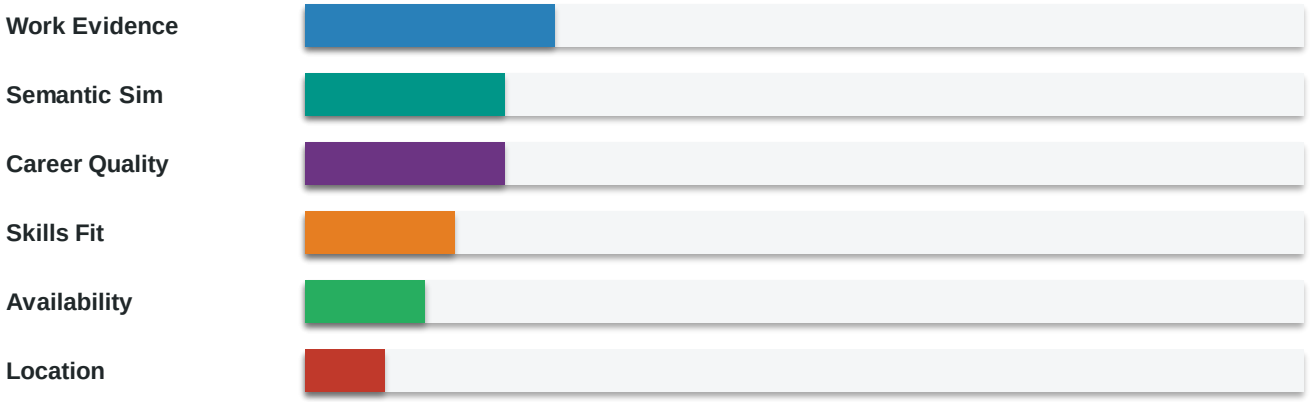
**Result:**
- Surfaces genuine builders regardless of how they wrote their skills section
- Phase 3: ranks 100K in 4.1s, CPU only
- **Zero API calls at rank time**

# JD Understanding & Candidate Evaluation

**Phase 0 — 1 GPT-4o-mini call extracts 3 artifacts from the job description**

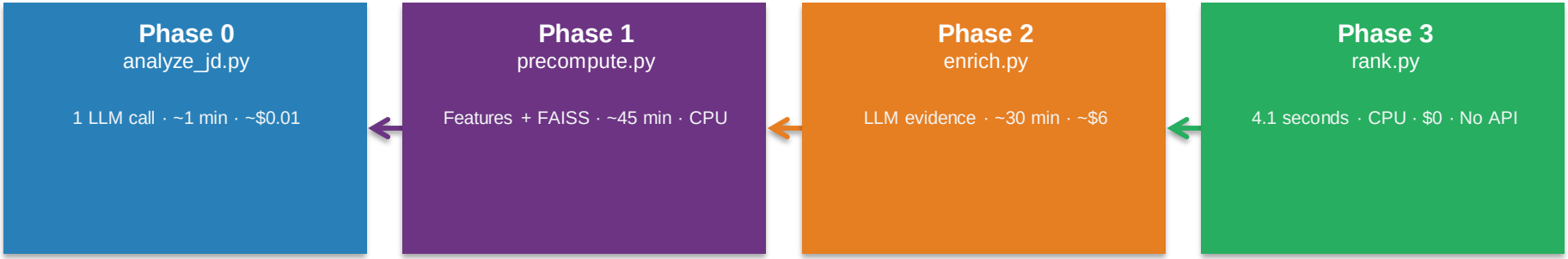
- production_experience_with_embeddings
- experience_with_vector_databases
- end_to_end_ml_systems
- startup_experience
- large_scale_ml_deployment
- llm_fine_tuning_or_rag

## Scoring Weights — How 6 Signals Combine into One Score




- 25% · LLM: did they actually build X in their career?
- 20% · FAISS cosine: work narrative vs JD query
- 20% · Product company history + experience band
- 15% · JD-weighted skill match + platform scores
- 12% · Open to work, response rate, notice period
- 8% · India / willing to relocate to Pune/Noida

# Ranking Methodology




## Final Scoring Formula

**final_score = (**

0.25 × work_evidence_score + 0.20 × semantic_similarity

+ 0.20 × career_quality_score + 0.15 × skills_fit_score

+ 0.12 × availability_score + 0.08 × location_score

**) × honeypot_multiplier**

× honeypot_multiplier → 0.0 kills fake profiles entirely

# Explainability & Data Validation

## Ranking Explainability

Top 200 candidates each get a 1-2 sentence recruiter note written by GPT-4o-mini.

**References real facts:**
- Actual career path
- Company type
- Confirmed work signals
- Relevant skills matched

Becomes the "reasoning" column in submission.csv

## Hallucination Prevention

**LLM reads career history ONLY**
Skills section excluded from prompt

**OpenAI strict JSON schema:**
- Forces boolean true/false only
- additionalProperties: false
- No free-text fabrication

**Prompt instruction:**
"If evidence absent or unclear, answer false"
On API failure: all-False fallback

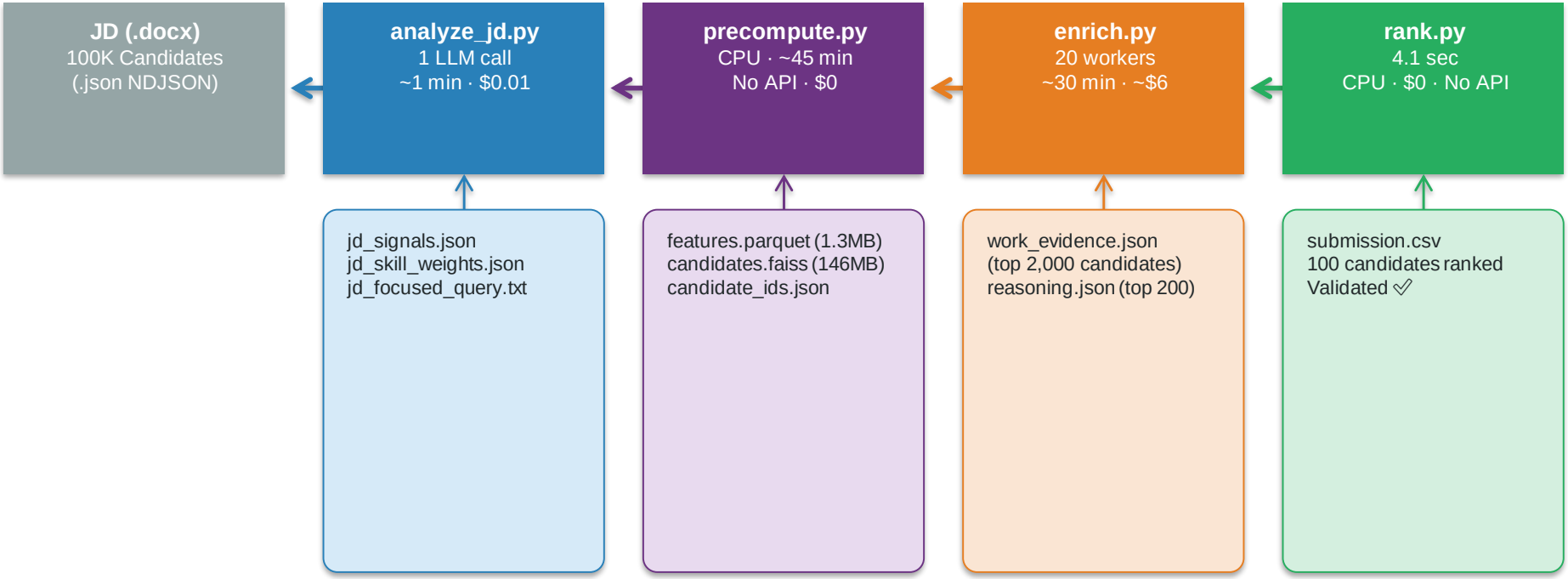
## Honeypot Detection

**5 rules fire before scoring**
2+ triggered → score = 0

① Declared exp > career length
② Overlapping full-time jobs
③ PhD before Bachelor's ends
④ Job title ≠ company domain
⑤ Skills contradict career

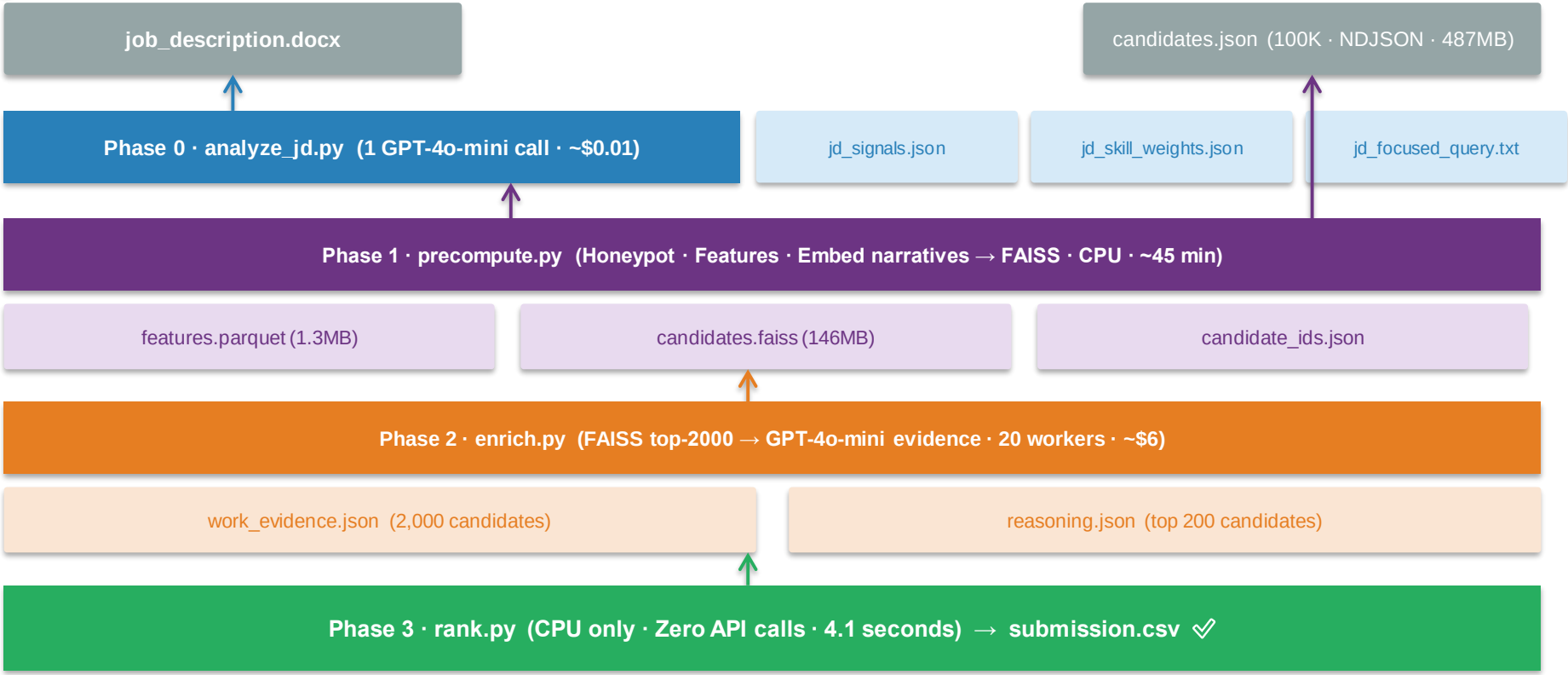
**Result:**
honeypot_multiplier = 0.0
**→ final_score = 0 (eliminated)**

# End-to-End Workflow




submission.csv

# System Architecture



# Results & Performance

**NDCG@10**

**1.0000**

Perfect — all top 10 are high-relevance

**NDCG@50**

**0.8527**

Strong broad ranking quality

**Phase 3 Runtime**

**4.1s**

of 300s budget (1.4% used)

**Honeypots in Top 100**

**0%**

Clean shortlist zero fake profiles

## All Hard Constraints Met

- ✓ Zero API calls at rank time — no internet needed
- ✓ CPU only — no GPU anywhere in the pipeline
- ✓ Exactly 100 rows, scores strictly non-increasing
- ✓ Passes official validate_submission.py with no errors
- ✓ Top score: 0.8631 · Score at rank 100: 0.5883
- ✓ 0 of 100 candidates are honeypots (0.0%)

# Technologies Used

**sentence-transformers**
**all-MiniLM-L6-v2**

Embeds work narratives into 384-dim vectors. CPU-friendly — chosen for speed/accuracy without any GPU requirement.

**faiss-cpu**
**IndexFlatIP**

Exact cosine similarity on 100K vectors in ~50ms on CPU. Deterministic— no approximation
errors. Zero GPU.

**openai · gpt-4o-mini**

Structured JSON (strict schema) for JD analysis + work evidence extraction.
Total pipeline cost: ~$6.60.

**pandas + pyarrow**

100K × 12 feature table stored as parquet (1.3MB, fast I/O). Typed columns prevent silent type errors.

**ThreadPoolExecutor**
**Python stdlib**

20 parallel LLM workers for evidence extraction. Reduced ~30 min sequential to ~3 min concurrent.

**streamlit**
**HuggingFace Spaces**

Live demo: CPU-only Docker container.
No secrets, no GPU, no setup for judges.
Auto-deploys on git push.

# Submission Assets

## GitHub Repository

**github.com/PrateekApurva/resume-shortlister**

Clean, complete working code
Phase branches: main / phase-1 / phase-2 / phase-3

**Reproduce from scratch:**
python analyze_jd.py
python precompute.py
python enrich.py
**python rank.py → 4.1 sec**

## HuggingFace Space (Live Demo)

**huggingface.co/spaces/PrateekApurva/resume-shortlister**

**Interactive Streamlit app:**
- Searchable top-100 candidate table
- Per-candidate score breakdown
- LLM work evidence (✓ / ✗)
- Full methodology explanation

Docker + Streamlit · CPU only
No GPU · No secrets needed

## Ranked Output: submission.csv

**100 candidates · Ranks 1–100**
Scores non-increasing

Top score: 0.8631
Score #100: 0.5883
**NDCG@10: 1.0000**
Honeypots: 0 (0.0%)

**Official validator:**
python validate_submission.py
**→ Submission is valid. ✓**

redrob | H2S

# INDIA.RUNS

Build what next India runs on

# THANK YOU